Final Project: Proposal

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1 Research Questions

This project focuses on residential housing price prediction and model reliability analysis in Metro Vancouver using multi-source urban, demographic, and macro-level data. The central objective is to assess the predictability of housing prices and to understand the reliability and limitations of predictive models across different neighborhoods. Within this shared analytical task, the project further examines how the same analysis can be conducted by human analysts and AI agent systems under identical objectives and evaluation criteria.

RQ1 is the primary data science objective (prediction, reliability, and bias analysis), while RQ2–RQ3 serve as secondary comparative questions to evaluate whether and how agentic workflows affect the quality and reliability of the same analysis under a controlled setting.

The core research questions are as follows:

1. How do multidimensional sociological factors, including policy orientation, ethnic composition, macro-economic data and age demographics, influence housing price predictions?

   Note: In this project, "policy orientation" will be operationalized through measurable *policy environment proxies* available in our datasets (e.g., zoning classification and taxation-related attributes in the Property Tax dataset, as well as broader market-policy context reflected by mortgage rates and rental market indicators), rather than subjective political stance.

   Therefore, we designed a hypothetical data analysis task:

   (a) How accurately can machine learning models predict residential housing prices in Metro Vancouver using multi-source urban and census data?

   (b) Do prediction errors systematically differ across neighborhoods with different socioeconomic characteristics (e.g., income level, immigration rate, housing type), and if so, to what extent?

   (c) Which factors contribute most to housing price predictions and model biases, and how do their impacts vary across different regions?

2. To what extent can an AI Agent system perform end-to-end data analysis tasks—from data cleaning and integration to modeling and forecasting—at a level comparable to human data analysts?

3. What are the key differences between AI-generated predictive models and traditional machine learning models developed by human analysts, and what underlying factors explain these differences?

2 Dataset Utilization

The datasets are organized into three analytical layers:
Layer 1 captures micro-level property attributes using the Property Tax Report.
Layer 2 represents the socio-demographic context, combining Census data for long-term neighborhood structure with IRCC statistics for short-term population inflows. We will additionally consider lagged versions (e.g., 3–12 month lags) of the inflow indicators to account for delayed market impacts.
Layer 3 reflects broader market and institutional conditions through mortgage lending rates and rental market indicators.

The study will integrate multiple public datasets to support comprehensive analysis:

- **Property Tax Report.csv [link]**
  Source: City of Vancouver Open Data Portal
  Brief description: This dataset contains detailed property-level information, including Property Identification Number (PID), legal type, zoning classification, land coordinates, year built, current land value, tax levy, and neighborhood code. These variables provide the foundation for modeling property valuation and spatial distribution.

- **Census Profile, 2021 Census of Population [link]**
  Source: Government of Canada
  Brief description: This dataset provides aggregated demographic and socioeconomic information, including age group distribution, ethnic and cultural origins, income levels, and language usage. These variables enable the analysis of social and demographic influences on housing prices.

- **Permanent Residents — Monthly IRCC Updates [link]**
  Source: Immigration, Refugees and Citizenship Canada (IRCC)
  Brief description: This dataset provides monthly aggregated counts of individuals granted permanent resident status in Canada, reported at regional levels. Due to privacy protection, values are rounded or suppressed for small counts. In this study, permanent resident admissions are used as a macro-level population inflow indicator to contextualize long-term housing demand dynamics, rather than as precise individual-level predictors. We will additionally consider lagged versions (e.g., 3–12 month lags) to capture potential delayed impacts on housing markets.

- **Study Permit Holders — Monthly IRCC Updates [link]**
  Source: Immigration, Refugees and Citizenship Canada (IRCC)
  Brief description: This dataset provides monthly aggregated statistics on temporary residents holding valid study permits. Similar to other IRCC datasets, small values are suppressed or rounded for privacy reasons. The data are used as a proxy for short-term population pressure and rental demand, and incorporated as contextual indicators in sensitivity and scenario-

based analyses. We will additionally consider lagged versions (e.g., 3–12 month lags) to capture potential delayed impacts.

- **Mortgage Lending Rate: 5-Year Fixed Term[link]**
  Source: Statistics Canada
  Brief description: This dataset provides time-series information on conventional mortgage lending rates, including the 5-year fixed-term mortgage rate, reported at regular temporal intervals. As a macro-level financial indicator, it captures broader monetary and credit conditions that may influence housing market dynamics. In this study, mortgage rate data will be used as a contextual variable for sensitivity analysis and scenario-based forecasting, rather than as a direct causal predictor.

- **Rental Market Report[link]**
  Source: Mortgage and Housing Corporation (CMHC)
  Brief description: This dataset provides aggregated indicators of rental market conditions in major Canadian cities, including vacancy rates and average rents. Reported at the city or metropolitan level, these indicators reflect supply-side constraints and rental market pressure within the housing system. Rental market data will be used as contextual variables to support sensitivity analysis and to help interpret model behavior and potential failure modes.

3 Methodology

3.1 Data Collection

Data will be collected by downloading the Property Tax dataset from the City of Vancouver Open Data Portal and the Census Profile dataset from the Government of Canada' s official website. All datasets will be stored in a centralized repository to ensure consistency across experimental groups.

3.2 Data Exploration

Exploratory Data Analysis (EDA) is necessary to understand data distributions, detect anomalies, and identify potential relationships.

Human Track: Human analysts will conduct EDA using visual inspection, descriptive statistics, and visualization tools such as histograms, scatter plots, and correlation matrices.

AI Track: Agent 1 (Data Engineer) will perform automated data profiling, generating statistical summaries, missing-value reports, and preliminary pattern detection using autonomous analysis scripts.

3.3 Data Cleaning

Data cleaning will focus on improving data quality and ensuring compatibility across sources.

- Handling missing values through imputation or removal.

- Standardizing column formats, data types, and naming conventions.

- Detecting and treating outliers.

Human Track: Manual cleaning using Pandas and custom scripts.

AI Track: Agent 1 will execute autonomous preprocessing pipelines based on predefined cleaning rules and adaptive heuristics.

3.4 Data Integration

The Property Tax dataset will be merged with Census data by mapping neighborhood codes to corresponding census tracts or geographic identifiers. Spatial and administrative boundaries will be harmonized to enable joint analysis at compatible aggregation levels.

3.5 Data Analysis

Both groups will conduct comprehensive quantitative and qualitative analyses. Mortgage lending rates will be incorporated as a time-aligned macroeconomic feature to capture borrowing cost dynamics and to support sensitivity analysis across different market conditions.

3.5.1 Train/Test Split and Validation

To avoid temporal leakage, we will adopt a time-aware split. Concretely, we will use historical data up to a cutoff year (e.g., data before 2024) for training and reserve the most recent period for testing. If sufficient time granularity is available, we will further use rolling/expanding-window validation within the training period for model selection and hyperparameter tuning.

3.5.2 Analysis Type

Macro-level population flow indicators, including permanent resident admissions and study permit holder counts, will be integrated as contextual demand-side features and analyzed for their differential impacts on prediction accuracy and error patterns across neighborhoods.

Human Track:

- Statistical analysis and correlation testing.
- Manual feature engineering, incorporating variables related to policy context, ethnicity, and age demographics.
- Machine learning model development using Scikit-learn and XGBoost, including regression and ensemble methods.
- Hyperparameter tuning and cross-validation.

AI Track: A multi-agent workflow consisting of:

- Agent 0 (Orchestrator / Human-in-the-loop): manages the workflow, decides whether to iterate (e.g., rerun cleaning, revise features, or retrain models) based on validation results and sanity checks, and enforces a fixed evaluation protocol to ensure comparability with the Human Track.
- Agent 1 (Data Engineer): Data preprocessing and structuring.
- Agent 2 (Multi-dimensional Analyst): Autonomous decomposition and analysis of complex social variables, including policy trends, ethnic composition, and demographic structure.
- Agent 3 (Demo/Predictor): Construction of predictive logic and interactive interfaces based on upstream outputs.

This layered architecture enables distributed reasoning and modular analysis.

3.5.3 Evaluation

The outputs of the Human Track and AI Track will be systematically compared using the following criteria:

- Prediction accuracy (e.g., MSE, RMSE, and trend alignment).
- Consistency of long-term price trends.
- Depth and interpretability of insights.
- Robustness across multiple experimental runs.

Additional subgroup analyses will examine whether model errors vary systematically during periods of elevated population inflow, as proxied by immigration and study permit statistics. Discrepancies will be analyzed to determine whether they arise from AI hallucination, modeling bias, or novel pattern discovery.

3.5.4 Data Product

The project will deliver:

- A comprehensive comparative research report documenting methodology, results, and interpretations.
- An interactive AI demonstration platform (Agent 3) capable of responding to user queries such as "Show me the 2027 housing price prediction" and visualizing forecast results.

4 Expected Impact

The primary expected impact of this study is to provide reliable and interpretable insights into housing price predictability in Metro Vancouver, including where predictive models succeed or fail and whether errors and uncertainty systematically differ across neighborhoods. Such findings can inform more responsible use of predictive analytics in urban decision-making and highlight potential community-level biases and limitations.

As a secondary contribution, the project empirically evaluates the feasibility of automating professional data analysis workflows using AI agents under a controlled and comparable evaluation protocol, clarifying when automation helps and when it may introduce reliability risks.

5 Potential Challenges

- Property tax data are recorded at the individual property level, while census data are aggregated at regional levels, leading to potential information loss and integration bias.
- AI agents may incorrectly infer causal relationships, such as attributing housing price changes to specific policies without sufficient evidence.
- Multi-agent systems may produce inconsistent results across different runs due to stochastic reasoning processes.

5.1 Theoretical Stance

- Reject the idea of a "perfect" prediction model.
- Shift focus from pure accuracy to model validity:
  - Identify failure conditions.
  - Detect potential structural or community-level biases.